作业（共4页，第1页）

- Bayes法则：
  1. 教材第一章 24；
  2. 【癌症诊断】一位医生按照“若病人有大于80%的概率患癌，则诊断病人需做手术”的原则为病人诊断。现有一位病人，根据前期检查，可判断有70%的概率患癌。医生请他进一步做了一个无创的影像检查，结果显示阳性。医生知道，根据以往经验，影响检查有20%的假阳性（即本来无癌症，但检查结果显示阳性），也有10%的假阴性（即本来有癌症，但检查结果显示阴性）。请问根据以上信息，医生现在是否应诊断病人需做手术?
  3. 请见下页

作业（共4页，第3页）

- Bayes法则（续）：

3. 【是失职，还是无辜？】2017年贵州航天医院三名从事尘肺病诊断的医生以“失职罪”被逮捕。公安机关调查2012-2016年3名医生共经手10708人胸片检查，针对诊断为尘肺病病人的1640份档案，对其中547例进行了重新复核，只查出42例有尘肺病。而据了解，一般和他们经验相当的一位医生独立读片的误判率为22%（即真实没有尘肺病、但却误判为尘肺病的概率）；在实际工作中，这三位医生独立地读片，当有两位或以上的医生认为有病就判为有病。因此当地警方认定他们对尘肺病的“误诊率超过90%”，“远超常规误判率”，造成3000万社保资金流失，以涉嫌“国有事业单位人员失职罪”予以羁押。

根据报道，这里我们可以合理假定 (1) 被复核的547份是随机抽取自1640份，因此该10708人的尘肺病发病率可以由 $\left(\frac{42}{547}\right) * 1640/10708$ 来估计 (注：可以用 比例 估计 概率 的严格概率支持要用到课程后续将学习的大数定律)；(2) 若真实有病，则三位医生读片时都能100%正确判定为有病。

到底是否失职？请你从概率的角度帮助判定！

作业（共4页，第4页）

- 事件的独立性：
  4. 教材第一章 33，
  5. （二选一）教材第一章 35，49，
  6. 随机抽查一个家庭，考察两个事件：A表示“至多有一个女孩”；B表示“男女孩子都有”。假设男女的出生率都是0.5。请证明：对3个孩子的家庭，A与B独立；而对4个孩子的家庭，A与B不独立。

7. 随机变量：

Let $\Omega = \{1,2,3,4,5,6\}$, and let $\mathcal{F} = \sigma(\{1,2,3,4\},\{3,4,5,6\})$. Is the function

$$X(\omega) = 2, \qquad \omega = 1,2,3,4$$

$$X(\omega) = 7, \qquad \omega = 5,6$$

a random variable over $(\Omega, \mathcal{F})$?

8. 随机变量的独立性：

Let X_1, X_2, X_3 be independent random variables with $P(X_i = 0) = P(X_i = 1) = \frac{1}{2}$. Let $A_1 = \{X_2 = X_3\}$, $A_2 = \{X_3 = X_1\}$, $A_3 = \{X_1 = X_2\}$. Prove that A_i’s are pairwise independent but not (totally) independent.

思考题

- 如何用公式表达和证明Bayes’ Rule的一致性?
- 举反例说明$P(\cap_{i=1}^{n} A_i) = \prod_{i=1}^{n} P(A_i)$不能保证$A_1, \dots, A_n$相互独立.

教材题目的具体内容

Section 1.4.2. Bayes' Rule

Problem 24. The prisoner's dilemma. The release of two out of three prisoners has been announced, but their identity is kept secret. One of the prisoners considers asking a friendly guard to tell him who is the prisoner other than himself that will be released, but hesitates based on the following rationale: at the prisoner's present state of knowledge, the probability of being released is 2/3, but after he knows the answer, the probability of being released will become 1/2, since there will be two prisoners (including himself) whose fate is unknown and exactly one of the two will be released. What is wrong with this line of reasoning?

Section 1.5. Independence

Problem 33. Using a biased coin to make an unbiased decision. Alice and Bob want to choose between the opera and the movies by tossing a fair coin. Unfortunately, the only available coin is biased (though the bias is not known exactly). How can they use the biased coin to make a decision so that either option (opera or the movies) is equally likely to be chosen?

Problem 35. Reliability of a k-out-of-n system. A system consists of n identical components, each of which is operational with probability p, independent of other components. The system is operational if at least k out of the n components are operational. What is the probability that the system is operational?

Problem 49. De Méré's puzzle. A six-sided die is rolled three times independently. Which is more likely: a sum of 11 or a sum of 12? (This question was posed by the French nobleman de Méré to his friend Pascal in the 17th century.)